

ERIEAU, ON, Canada

WMO: 714650

Lat: 42.257N

Lon: 81.908W

Elev: 178

StdP: 99.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-13.0	-20.5	0.6	-15.0	-17.6	0.8	-12.2	15.8	-5.6	14.2	-5.3	4.8	270	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	27.0	22.9	26.0	22.3	25.1	21.7	24.5	25.7	23.6	25.0	22.8	24.4	3.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.5	25.3	23.2	18.3	24.6	22.3	17.3	24.0	75.2	25.9	71.7	25.2	68.3	24.4	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	11.0	9.6	DB	-18.1	29.6	3.6	1.7	-20.7	30.8	-22.8	31.8	-24.8	32.8	-27.4	34.0
			WB	-18.6	25.1	3.4	1.5	-21.1	26.1	-23.1	27.0	-25.0	27.8	-27.5	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-3.7	-3.4	0.4	6.4	12.4	18.6	21.5	21.6	18.4	12.2	6.1	0.5
	DBStd	9.86	5.10	4.80	4.08	3.24	3.33	3.08	2.45	2.33	3.38	3.92	4.41	4.53
	HDD10.0	1672	425	376	296	114	13	0	0	0	0	22	129	296
	HDD18.3	3587	683	609	554	359	188	34	3	3	39	191	368	554
	CDD10.0	1422	0	0	0	5	87	257	358	359	253	92	12	0
	CDD18.3	296	0	0	0	0	3	42	103	104	42	3	0	0
	CDH23.3	786	0	0	0	0	6	68	327	312	73	0	0	0
	CDH26.7	43	0	0	0	0	1	6	23	12	1	0	0	0
Wind	WSAvg	4.3	5.5	4.8	4.5	4.7	4.0	3.2	3.0	3.1	3.7	4.6	5.5	5.4
Precipitation	PrecAvg	917	75	62	61	76	84	79	85	74	81	77	81	75
	PrecMax	1200	118	125	123	125	152	158	164	133	214	154	148	147
	PrecMin	720	30	17	22	26	23	31	31	13	18	20	30	31
	PrecStd	142	22	27	29	26	33	33	29	28	45	32	33	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.2	9.9	15.7	23.0	27.2	28.8	27.7	26.7	21.7	15.7	11.1
		MCWB	6.3	4.8	6.8	10.3	18.1	21.1	23.8	23.1	23.2	19.8	13.9	10.4
	2%	DB	5.6	4.5	7.9	12.9	20.2	25.1	27.1	26.8	25.2	20.3	14.4	9.5
		MCWB	4.7	3.2	5.9	8.9	16.3	21.0	23.3	22.9	22.1	17.9	12.6	8.6
	5%	DB	4.3	3.3	6.4	11.3	18.1	23.8	26.1	25.8	24.1	19.0	13.3	7.9
		MCWB	3.2	2.1	4.6	8.2	15.1	20.4	22.7	22.1	21.1	16.7	11.3	6.7
	10%	DB	2.8	2.2	5.2	10.2	16.6	22.7	25.1	25.0	23.1	17.8	12.3	6.3
		MCWB	1.8	1.1	3.7	7.5	13.8	19.8	21.8	21.8	20.3	15.3	10.5	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	5.3	8.0	12.2	19.0	23.8	25.8	25.7	24.2	20.6	14.7	10.6
		MCDB	6.9	5.8	8.9	14.2	21.4	25.1	27.1	26.5	25.6	21.4	15.4	10.9
	2%	WB	4.9	3.5	6.2	10.0	17.1	22.3	24.7	24.5	23.1	18.2	13.1	8.8
		MCDB	5.5	4.2	7.4	11.6	19.5	23.9	26.0	25.5	24.6	19.7	13.9	9.3
	5%	WB	3.3	2.3	4.9	8.9	15.6	21.4	23.7	23.6	22.1	17.2	11.8	6.9
		MCDB	4.0	3.1	6.0	10.6	17.7	23.1	25.1	25.0	23.6	18.7	12.9	7.9
	10%	WB	1.9	1.3	3.8	7.8	14.5	20.4	22.8	22.7	20.8	15.9	10.6	5.1
		MCDB	2.7	2.1	5.1	9.5	16.1	22.2	24.4	24.4	22.7	17.5	12.1	6.1
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.3	5.0	5.3	5.1	4.9	5.1	4.9	5.0	5.2	4.8	4.5
		MCDBR	4.8	5.1	6.1	7.2	7.0	6.0	5.6	5.1	4.7	4.8	5.1	4.5
		MCWBR	4.9	4.7	4.8	5.0	4.5	3.9	4.1	4.1	3.9	4.4	5.1	4.8
	5% WB	MCDBR	4.6	5.0	5.8	6.6	6.2	4.7	4.8	4.4	4.4	4.6	4.9	4.6
		MCWBR	4.9	4.8	5.0	5.4	4.7	3.7	4.0	4.5	4.2	4.8	5.2	4.9
Clear-Sky Solar Irradiance	taub	0.304	0.327	0.347	0.395	0.421	0.440	0.436	0.413	0.380	0.366	0.327	0.305	
	taud	2.392	2.286	2.315	2.267	2.245	2.232	2.270	2.321	2.394	2.403	2.475	2.480	
	Ebn at Noon	837	869	895	873	856	838	837	846	851	817	806	809	
	Edh at Noon	80	104	114	129	136	138	132	121	105	92	74	68	
All-Sky Solar Radiation	RadAvg	1.45	2.11	3.45	4.59	5.49	6.20	6.16	5.39	4.30	2.74	1.68	1.16	
	RadStd	0.14	0.29	0.35	0.41	0.47	0.41	0.37	0.25	0.37	0.23	0.24	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	N/A

Nomenclature: See separate page